

Model Risk Management Documentation

Flight Departure Delay Prediction Model

American Airlines | Charlotte Douglas International Airport (CLT)

DSBA 6156 | Matthew Sweeney | April 2026

1. Model Overview

1.1 Model Name

Flight Departure Delay Prediction Model

1.2 Model ID / Version

v1.0 - Initial Release

1.3 Date Created / Last Updated

April 2026

1.4 Model Owner

Matthew Sweeney (DSBA 6156 Final Project)

1.5 Model Developer(s)

Matthew Sweeney - DSBA 6156, UNC Charlotte

1.6 Model Reviewer(s)

Course Instructor - DSBA 6156

1.7 Model Users / Stakeholders

- Primary: Air travelers booking flights out of Charlotte Douglas International Airport (CLT)

1.8 Purpose & Business Use Case

This model predicts whether an American Airlines flight departing from Charlotte Douglas International Airport (CLT) will be delayed by more than 15 minutes. The goal is to help passengers make better decisions when booking flights by estimating the likelihood of a delay.

Passengers can use this information to choose flights with lower delay risk or allow more time for connections. This is especially useful in Charlotte, where many travelers rely on connecting flights.

1.9 Model Category / Type

Supervised Machine Learning - Binary Classification (Gradient Boosting Classifier)

1.10 Criticality / Risk Tiering

Risk Tier: Low-Medium

Rationale: This model is designed to help passengers make more informed decisions, but it does not directly control any systems or operations. If the predictions are off, the main impact would be a less optimal travel choice rather than anything involving safety or financial risk.

This model is intended as a decision-support tool rather than a fully automated system. While it uses a more advanced method (Gradient Boosting), any mistakes it makes are unlikely to have a significant real-world impact.

2. Model Scope & Boundaries

2.1 In-Scope Components

- Data ingestion and cleaning from BTS Excel files
- Feature engineering (e.g., departure hour, day of week, destination airport, scheduled flight duration)
- Creation of binary target variable (delay > 15 minutes = 1)
- Model training, validation, and comparison across logistic regression, random forest, and gradient boosting
- Feature importance analysis
- Batch prediction generation for scheduled flights

2.2 Out-of-Scope Components

- Real-time data inputs (e.g., weather, air traffic control)
- Flight itinerary optimization or routing decisions
- Post-departure delay tracking or live updates
- Automated decision-making (e.g., ticketing, rebooking, pricing)
- Flights operated by airlines other than American Airlines

2.3 Dependencies & Interfaces

- Data Source: U.S. Bureau of Transportation Statistics (BTS) - Detailed Statistics on Departures (Excel format)
- End Users: Passenger-facing booking or travel planning interfaces where delay risk is displayed during flight selection (batch integration)
- Tools & Platform: Python (scikit-learn, pandas, matplotlib); developed in Google Colab

3. Data Documentation

3.1 Data Sources

Attribute	Detail
Source Name	U.S. Bureau of Transportation Statistics (BTS)
Dataset	Detailed Statistics on Departures - American Airlines, CLT
Coverage Period	2024-2025
Format	Excel (.xlsx files)
Access	Publicly available at bts.gov
Refresh Frequency	Monthly (historical batch)
Owner	U.S. Department of Transportation

3.2 Data Quality Assessment

- **Completeness:** The raw file contained header rows requiring the first 7 rows to be skipped. Unnamed columns were dropped. Missing values in key feature columns were removed by removing those rows.
- **Accuracy:** Data originates from official carrier-reported BTS records, which are widely used for operational analysis.
- **Timeliness:** Data covers 2024-2025 and reflects recent operational patterns.
- **Consistency:** Departure time field required cleaning to remove non-numeric characters and standardize to a 4-digit HHMM format before hour extraction.
- **Known Issues:** A small number of rows contained invalid departure time values and were dropped. Very early hour bins (midnight-2am) showed unusually high delay rates likely due to low flight volume, which may introduce noise into the model.

3.3 Data Preprocessing

- Loaded the Excel dataset and skipped the first 7 rows to remove the BTS header metadata
- Removed unnecessary columns (e.g., unnamed columns)
- Converted date fields to datetime format and extracted month and day-of-week features
- Cleaned the scheduled departure time by removing non-numeric characters, standardizing to a 4-digit format, and extracting the hour
- Created a binary target variable where Delayed = 1 if departure delay > 15 minutes, and 0 otherwise
- Applied one-hot encoding to categorical features such as day of the week and destination airport

- Removed rows with missing values in any modeling feature

3.4 Training / Testing / Validation Samples

- Data was split into training and testing sets using a random 80/20 split
- 80% of the data was used for training and 20% was held out for evaluation
- A fixed random state (42) was used to ensure reproducibility
- The dataset covers flight operations from 2024 to 2025
- The data is imbalanced, with more on-time flights than delayed flights; this was addressed using class weighting and sample weighting techniques

4. Methodology & Model Design

4.1 Conceptual Design

The model treats each scheduled flight departure as an independent observation. Using features available at the time of booking (departure hour, day of week, destination airport, and scheduled flight duration), the model outputs a binary prediction of whether the flight will be delayed by more than 15 minutes. The 15-minute threshold aligns with the standard industry definition used by the FAA and BTS.

4.2 Algorithm Selection Rationale

Three algorithms were trained and evaluated:

- Logistic Regression (baseline): A simple and interpretable linear classifier used as a baseline. Class imbalance was addressed using class weighting. While the model is easy to interpret, it showed lower performance in identifying delayed flights compared to the other models.
- Random Forest: An ensemble method capable of capturing non-linear relationships and feature interactions. Class weighting was applied to address imbalance. The model achieved moderate recall (0.46) for delayed flights, improving over logistic regression but still underperforming compared to Gradient Boosting.
- Gradient Boosting (selected model): A sequential ensemble method that improves performance by correcting errors from previous models. Sample weighting was used to handle class imbalance. This model achieved the highest recall (0.70) and F1 score for delayed flights, making it the most effective choice for the primary use case.

Gradient Boosting was selected as the final model because the goal of the project is to identify flights at risk of delay, and this model was most effective at capturing delayed flights after addressing class imbalance.

4.3 Model Architecture

- Algorithm: Gradient Boosting Classifier (scikit-learn implementation)
- Number of estimators: 100
- Learning rate: 0.1
- Max depth: 3
- Random state: 42 (for reproducibility)
- Class imbalance handling: Addressed using sample weighting to give higher importance to delayed flights during training.

4.4 Feature Engineering

Feature	Description	Type	Rationale
Hour	Departure hour (0-23) extracted from scheduled departure time	Numeric	Delays tend to accumulate throughout the day, making departure hour an important predictor
Day of Week	Day of week (Monday-Sunday), one-hot encoded	Categorical	Travel patterns vary by day and may influence delay risk
Destination Airport	IATA code of destination, one-hot encoded	Categorical	Destination-specific congestion affects delay rates
Scheduled elapsed time (Minutes)	Planned flight duration in minutes	Numeric	Longer flights may face greater exposure to disruptions

4.5 Model Assumptions

- Statistical: The training data is expected to represent future flight activity from CLT. Each flight is treated as an independent observation.
- Business: The 15-minute delay threshold follows the standard industry definition and is appropriate for passenger decision-making.
- Data: BTS-reported delay values are assumed to be accurate and complete. Scheduled flight times are assumed to be available at the time of booking.
- The model assumes no major external disruptions (e.g., severe weather events or system-wide outages)

4.6 Limitations

- Technical: The model does not incorporate weather, air traffic control, or mechanical delay causes, which are major drivers of real-world delays.
- Data: The dataset is limited to a single carrier (American Airlines) and a single origin airport (CLT). Predictions are not applicable to other airports or carriers.
- Business: The model outputs delay probability based on historical patterns and cannot account for unusual events such as weather systems, strikes, or infrastructure outages.
- Ethical/Fairness: The model does not use personal or sensitive data. However, some features, such as destination airport, may still indirectly reflect broader patterns. A formal bias analysis was not conducted.

5. Model Performance

5.1 Training Performance Metrics

The model was evaluated on a held-out 20% test set. The following results reflect performance after applying class balancing:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.597	0.355	0.641	0.457
Random Forest	0.633	0.352	0.459	0.398
Gradient Boosting	0.571	0.346	0.701	0.464

Recall is particularly important in this context, as the goal is to identify flights at risk of delay. Gradient Boosting achieved the highest recall (0.701), making it the most effective model for this use case. While Gradient Boosting has slightly lower overall accuracy and precision compared to some models, this tradeoff is acceptable because the primary goal is to identify delayed flights rather than minimize false positives.

5.2 Validation Performance

The test set (20% holdout, random split) was used as the primary validation dataset. No additional external dataset was used for validation. Gradient Boosting achieved the highest recall (0.70) and F1 score among all three models evaluated.

5.3 Backtesting

Backtesting was not performed, as the model was evaluated using a standard train/test split on historical data. A time-based validation approach could be explored in the future to assess performance across different periods.

5.4 Sensitivity Analysis

Formal sensitivity analysis was not conducted. However, insights from exploratory data analysis provide useful indicators of model sensitivity:

- Delay rates increase throughout the day, suggesting the Hour feature is a strong and consistent predictor.
- Very early morning hours (low flight volume) show irregular delay patterns, which may reduce prediction reliability for that segment.
- Feature importance analysis from the Gradient Boosting model confirms that departure hour and scheduled flight duration are the most influential predictors.

5.5 Benchmarking

Three models were compared during development (see section 4.2). Gradient Boosting outperformed both Logistic Regression and Random Forest in identifying delayed flights, achieving the highest recall.

No external benchmark datasets or industry model comparisons were used.

6. Explainability & Interpretability

6.1 Global Explainability

Feature importance from the Gradient Boosting model was used to understand which variables have the most influence on delay predictions. The top predictors identified are:

- Departure Hour - the strongest predictor; delays tend to increase later in the day
- Scheduled Elapsed Time (Minutes) - longer flights are more likely to experience delays
- Day of Week - moderate impact, with some variation across days
- Destination Airport – smaller overall impact, with only minor differences across locations

No external explainability tools (such as SHAP or LIME) were used. These could be added in the future to provide more detailed, prediction-level explanations. Overall, the results are consistent with earlier exploratory analysis and support the model's behavior.

6.2 Local Explainability

The model can be used to generate predictions for individual flights, including an estimated probability of delays. For example, a flight departing at 9pm with a 3-hour duration would likely receive a high delay risk score based on the learned patterns. However, predictions may be less reliable in edge cases, such as very early morning flights, where the data is limited.

6.3 Business Interpretability Assessment

The main drivers of delay risk (time of day and flight duration) are easy to understand and explain to non-technical users. For instance, evening flights generally have a higher likelihood of delays compared to morning departures. Overall, the model is interpretable and appropriate for this type of use case, where it is intended to support (not replace) decision-making.

7. Fairness, Bias, and Ethical Assessment

7.1 Protected Attributes Reviewed

The model does not use any protected demographic attributes (such as race, gender, age, or income). All input features are based on operational flight characteristics available in public BTS data.

7.2 Bias Metrics Evaluated

No formal bias metrics were computed, as the model does not make decisions about individuals based on personal characteristics. The model produces delay risk scores for flights, and these outputs are equally available to all users querying the same flight.

7.3 Fairness Mitigation Approaches

Class imbalance between delayed and on-time flights was handled using sample weighting to reduce bias toward the majority (on-time) class. This allows the model to better account for delayed flights during training.

7.4 Ethical Considerations

- The model is intended to support decision-making and does not directly restrict or penalize any passengers or airline carriers.
- Predictions are based on historical data and results should be presented with some level of uncertainty.
- No personally identifiable information (PII) is used in building or applying the model.
- The model should not be used to support unfair pricing or differences in how passengers are treated.

8. Model Controls & Governance

8.1 Access & Permission Controls

- Model training code is stored in a Google Colab notebook with access limited to the developer.
- Trained model artifacts are stored in a controlled environment with access restricted to authorized systems.
- End users only see model outputs (delay risk scores), not the underlying model or training data.

8.2 Change Management

- Model code changes are tracked using Google Colab revision history.
- Any updates to features, hyperparameters, or training data require re-evaluation on the test set before deployment.
- Model versions are tracked in the Model ID field (currently v1.0).

8.3 Model Inventory Registration

This documentation serves as the primary model record for version 1.0. Future updates should include a new Model ID and update relevant sections of this document.

9. Model Monitoring Strategy

9.1 Performance Monitoring

Metric	Threshold	Frequency
Recall (Delayed class)	Alert if drops below 0.55	Monthly
F1 Score	Alert if drops below baseline level	Monthly
Prediction distribution	Alert if the percentage of predicted delays shifts by more than 10 points	Monthly

9.2 Data Drift & Stability Monitoring

- Feature drift: Monitor the distribution of Hour and Scheduled Elapsed Time in incoming data monthly. Significant shifts (e.g., schedule changes or new routes) may require model retraining.
- Label drift: Monitor actual delay rates in production data monthly. If the base rate of delays changes significantly from the training distribution, recalibration may be needed.
- Recommended tool: Population Stability Index (PSI) for key features

9.3 Alerts & Escalation Procedures

- If performance metrics fall below defined thresholds, the model developer should be notified and retraining should be initiated.
- If the data source structure changes (e.g., BTS export format updates), the data pipeline must be updated before generating new predictions.
- Escalation path: Model Developer -> Course Instructor / Model Owner

9.4 Periodic Review Requirements

- Quarterly: Review model performance metrics against recent flight data.
- Annually: Conduct full model revalidation, including retraining on updated BTS data, re-evaluating feature importance, and reassessing class imbalance handling.
- Trigger-based: Retraining is required following major operational changes (e.g., new airline routes, schedule restructuring, or major infrastructure changes at CLT).

12. Documentation Appendix

Glossary

Term	Definition
BTS	Bureau of Transportation Statistics - U.S. federal agency that collects and publishes aviation performance data
CLT	Charlotte Douglas International Airport (IATA code)
Delay	Departure delay exceeding 15 minutes, based on FAA/BTS standard definitions
Recall	Proportion of actual delayed flights correctly identified by the model
Precision	Proportion of predicted delays that are actual delays
F1 Score	Harmonic mean of precision and recall; balancing both metrics
Class Imbalance	Condition where one class (on-time) significantly outnumbers another (delayed) in the dataset
Sample Weight	Per-observation weight assigned during training to compensate for class imbalance
GBM	Gradient Boosting Machine - ensemble learning method that builds sequential decision trees to improve model performance
PSI	Population Stability Index - metric for detecting feature distribution drift over time

Code Reference

Key modeling code is maintained in: `Final_Project_DSBA_6156.ipynb`

Primary libraries used: pandas, numpy, scikit-learn, matplotlib, seaborn

The final model is implemented using a Gradient Boosting classifier, with sample weighting applied to address class imbalance.

Data Source Reference

U.S. Bureau of Transportation Statistics - Airline On-Time Statistics

<https://www.transtats.bts.gov>

Dataset: Detailed Statistics on Departures (American Airlines, Charlotte Douglas International Airport - CLT), 2024-2025